Exploiting the Overnight Risk Premium

Cross-Sectional Evidence from US Equity Markets

CONCRETUM RESEARCH
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"Money never sleeps" is one of the most famous expressions on Wall Street. Yet decades of equity market data suggest a curious twist: **money** may never sleep, but it appears to **do most of its work at night**.

One of the strangest facts in quantitative finance is that **investors** have historically **earned** a **disproportionately** large share of their long-term equity **returns outside regular trading hours**. In other words, much of the reward for owning stocks appears to have been generated between the close of one session and the open of the next.

As unusual as it may sound, this behavior happens to be one of the most widely **documented stylized facts** in **quantitative finance**, with evidence dating back to the influential work of *Cooper, Cliff* and *Gulen* (2008), who showed the effect to hold not only for individual stocks but for equity indices and index futures alike.

Seeing that many readers appreciated our last series of articles, in which we investigated overnight opportunities across stocks that are frequently targeted by active intraday short sellers, we thought it would be a good moment to present some **additional findings** around the *overnight risk premium*.

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In particular, today we focus on some of the cross-sectional features that we find to be most predictive of positive overnight returns.

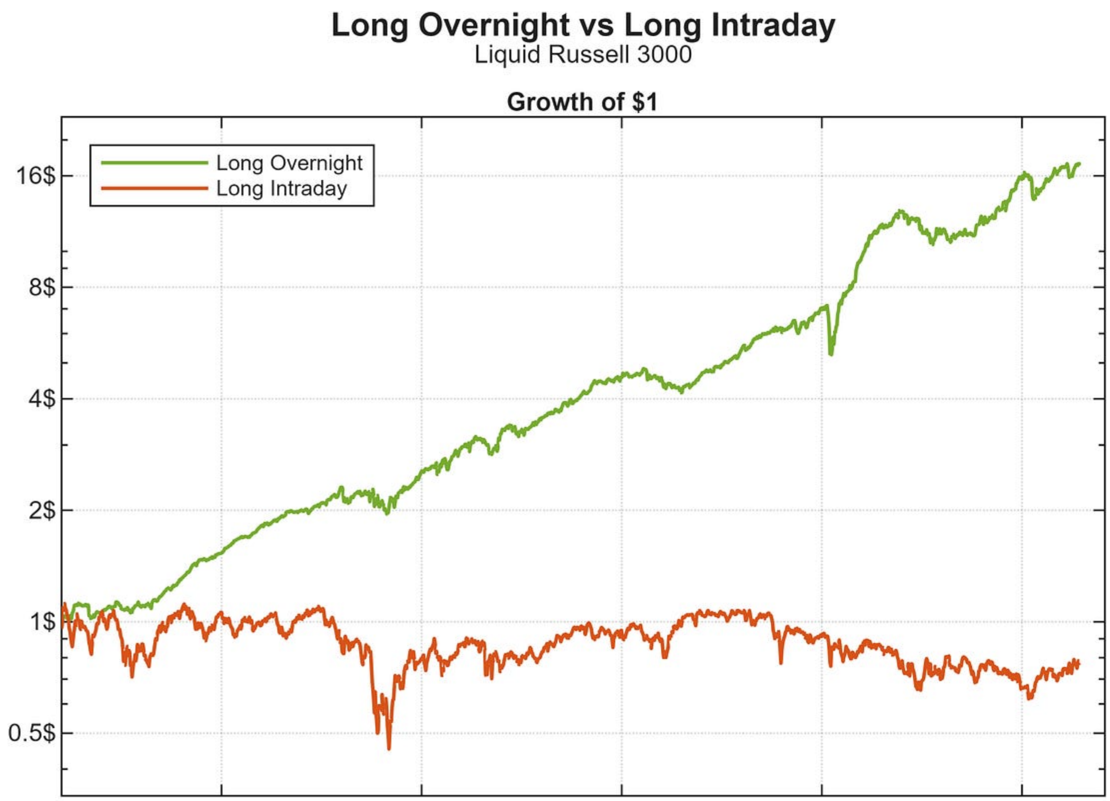
Is the Overnight Risk Premium Dead?

The first thing we do, is to assess whether market participants have indeed continued to be compensated for holding equity exposure overnight rather than intraday.

To do so, we construct **two distinct equal-weighted benchmark portfolios** that trade across a wide basket of Russell 3000 names. To ensure a **minimum standard of liquidity**, we restrict activity to the 2,000 most liquid securities within the index, where liquidity is measured as the median dollar volume exchanged on a rolling 252-day basis: we refer to this universe throughout as the *Liquid Russell 3000*.

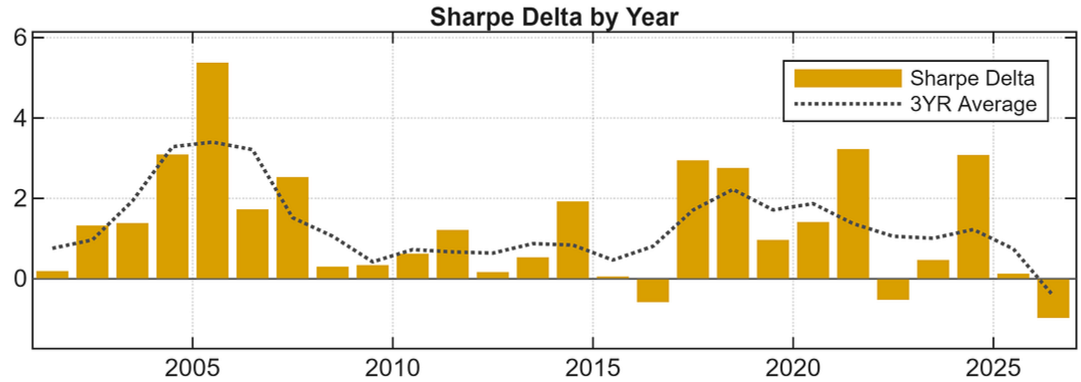
The first benchmark portfolio simply buys every tradable stock at the open of day *t* and **liquidates all positions at the close** of the same session; the second buys at the close of day *t*, carries the positions overnight, and **sells into the following open** (*t+1*): importantly, both portfolios draw from the same subset of Russell 3000 names, with eligibility evaluated at the close of day *t-1*.

Even a visual inspection of the chart below reveals a clear divergence between the two portfolios. While the **overnight strategy compounds steadily throughout the sample, the intraday portfolio struggles to generate meaningful growth**, suggesting that a substantial portion of the equity risk premium has historically been earned outside regular trading hours.



For every calendar year in our sample, we calculate the **Sharpe ratio of the overnight portfolio and compare it against the Sharpe ratio of the intraday portfolio:** we focus on the difference between the two rather than on raw returns because the risk profile of overnight and intraday strategies is not constant through time.

Measuring excess performance on a risk-adjusted basis allows us to isolate whether the overnight window delivered superior compensation per unit of risk, rather than simply benefiting from periods of elevated market volatility.



Looking at a rolling three-year average of the spread, the **overnight-minus-intraday dynamic** appears to have been **strongest** in the years leading up to the publication of Cooper, Cliff and Gulen (2008). Since then, the **effect has remained visible, although generally with less intensity**.

In recent years (2022–June 2026), the effect appears somewhat weaker still, owing in large part to **2024 having been the only year in that stretch to deliver a strongly positive spread** in the risk-adjusted returns of the two benchmark portfolios.

Judging from our universe, then, even though the overnight-minus-intraday dynamic may no longer be as persistent and clean as it was in the early 2000s, (something we might reasonably have expected) it remains clearly observable in recent years, and we would say it is **far from dead or "arbitraged away"**.

Predicting Overnight Returns

We now move to a more advanced step, one that goes beyond simply documenting the existence of the overnight risk premium.

Rather than searching for entirely new market anomalies, we ask a different question:

What makes an existing anomaly stronger?

In other words, once a persistent market behavior has been identified, can we find **features** that **help explain when it is likely to be more pronounced** and when it is likely to be weaker?

We refer to these characteristics as *satellite effects*. A satellite effect is not a standalone anomaly in its own right, but rather a feature that enhances, amplifies, or helps explain the behavior of another anomaly.

In our case, the anomaly of interest is the overnight drift in equities: the objective is therefore to identify stock characteristics that may help predict where that drift is most likely to occur.

With that goal in mind, we select the following features, some rooted in traditional factor investing and others more directly connected to the nature of the overnight return phenomenon itself. Our expectation is that these characteristics should contain **information** about **subsequent overnight returns.**

Feature	Description
252D Median $ Volume	Median daily dollar volume over the past year (our proxy for how liquid a name is)
252D Vol (Close to Close)	Volatility of close-to-close returns over the past year
252D Vol (Overnight Moves)	Volatility of close-to-open moves over the past year
Prior Overnight Move	Last night's gap (this morning's open against the prior close)
Prior Close $ Price	Yesterday's close in dollars, unadjusted for splits and dividends
252D Return	Twelve-month price return, our momentum measure with no skip period
21D Return	One-month price return, often used in reversal-type factors
252D Skew	Asymmetry of daily close-to-close returns over the past year
21D MAX Overnight Move	Largest single overnight gain over the past month

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Note - Importantly, we construct each of these rolling features in a trade-aware fashion, so that all of them are already known by the open of day t, leaving a few hours of lag before actual execution at the same session close.

What we then do is straightforward: **on a daily basis, we sort the full Liquid Russell 3000 by each feature into ten decile portfolios**, from D1 (the lowest feature values) to D10 (with the highest values of each feature), and take each decile’s average overnight return.

Finally, for each decile portfolio, **we compute full-sample Sharpe ratios**: all values are presented in the table below.

Feature	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
252D Median $ Volume	1.12	1.19	1.14	0.97	0.94	0.91	0.87	0.85	0.83	0.78
252D Vol (Close to Close)	0.64	0.51	0.46	0.55	0.56	0.63	0.63	0.81	1.20	2.24
252D Vol (Overnight Moves)	0.70	0.58	0.55	0.60	0.60	0.64	0.71	0.90	1.19	2.09
Prior Overnight Move	0.48	0.44	0.50	0.58	0.63	0.74	0.86	0.95	1.28	2.52
Prior Close $ Price	2.56	0.97	0.76	0.67	0.63	0.60	0.57	0.59	0.65	0.66
252D Return	2.01	0.73	0.48	0.44	0.48	0.59	0.66	0.80	1.01	1.59
21D Return	2.17	1.05	0.78	0.70	0.60	0.58	0.58	0.64	0.67	1.02
252D Skew	1.22	0.93	0.92	0.95	0.91	0.86	0.88	0.90	0.93	1.15
21D MAX Overnight Move	0.06	0.33	0.39	0.52	0.61	0.79	0.91	1.12	1.39	2.09

All things considered, the pattern appears quite clear: it is the most volatile, the most illiquid and the lowest-priced names that have historically earned the highest (gross-of-costs) risk-adjusted overnight returns, with several deciles posting **Sharpe ratios above 2.**

Several patterns emerge from the table.

When stocks are sorted by liquidity, the **least liquid names** tend to **earn higher overnight returns** than their more heavily traded counterparts. A similarly strong relationship appears when sorting by stock price, with **lower-priced securities exhibiting substantially stronger overnight performance** than higher-priced names.

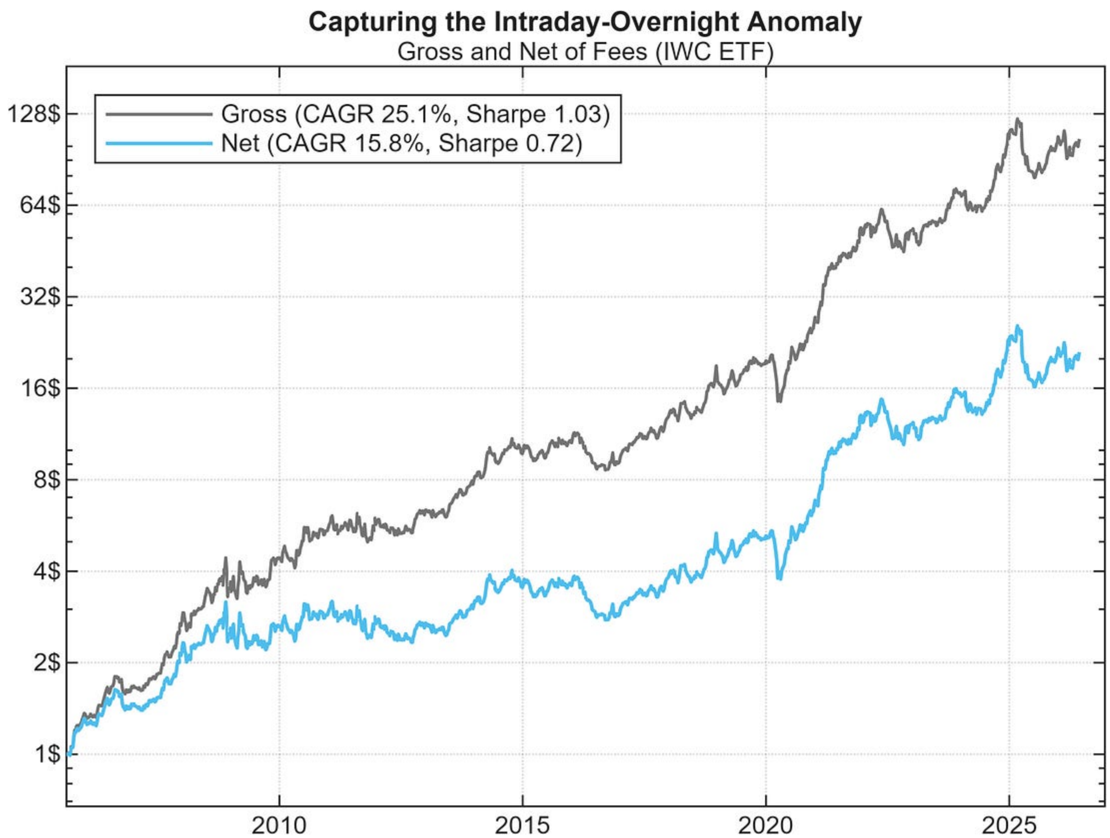
Perhaps even more interesting are the results associated with volatility and repricing activity. Stocks characterized by **elevated historical**

volatility, **large prior overnight moves**, or **unusually large overnight jumps** over the recent past consistently exhibit some of the **highest risk-adjusted overnight returns** in our sample.

At the same time, the results raise an obvious practical question. Many of the characteristics associated with the strongest overnight effects, including low liquidity, low price and elevated volatility, are also associated with **higher implementation costs** and greater execution complexity. This naturally leads us to ask whether **investors can gain exposure** to the **same phenomenon** through a **more accessible vehicle**, rather than trading dozens or hundreds of individual securities directly.

To explore this idea, we turn to the **iShares Micro-Cap ETF (IWC)**, a liquid instrument that provides diversified exposure to the smallest segment of the US equity market.

We test a “always-on” *long-overnight*, *short-intraday* signal (traded at a fixed ±100% exposure), and the results are presented below.



Portfolios assume trading of fractional shares. Fees are modeled using IBKR standard tier of $0.0035 / Share, with no minimum commission threshold (fee rate is doubled for sell transactions to conservatively account for additional sell-side charges)

Seen through the lens of the features we selected, the findings are perhaps not entirely surprising. By construction, **IWC provides broad exposure to smaller, lower-liquidity and generally more volatile companies**, precisely the segments where our earlier analysis identified the **strongest overnight return dynamics**.

More interestingly, the average P&L / Share (per single trade-leg) of ~$0.048 appears **sufficiently large to survive conservative trading-fees assumptions**: once those are included (see figure caption for details), CAGR declines by ~930 bps, but the overall profitability of the signal remains intact.

It is worth noting, however, that these estimates do not incorporate market impact (i.e. slippage) or borrowing costs. While such frictions may be modest for smaller allocations, they can become increasingly relevant as capital scales. As a result, actual realized performance could be lower than the figures reported here.

Even so, the overall picture remains encouraging: **a meaningful fraction of the overnight-versus-intraday spread observed in individual micro-cap stocks appears to survive aggregation into a liquid and operationally simple ETF**.

Practical Takeaways

For our readers, we believe there are **two** particularly **interesting takeaways** from this analysis.

The **first** relates to **stock selection**.

The **overnight premium** does not appear to be distributed uniformly across the equity universe. Instead, it seems to **concentrate among stocks** that share a number of common characteristics, including **elevated volatility**, **lower liquidity** and **lower absolute price** levels. Perhaps even more interestingly, some of the strongest results emerge among stocks that have recently experienced unusually **large overnight repricing events**.

This naturally raises the question of whether **combining several of these features** can further improve the signal and help **identify** those **securities** most likely to experience **unusually strong overnight returns**.

The **second takeaway** relates to a concept we discussed earlier this year in our article on ***Fast Alphas.***

Recall that a **fast alpha is not necessarily a standalone strategy**, but rather a **short-lived** source of **edge** that can be used to improve the implementation of an existing investment process.

Viewed through this lens, the **overnight-versus-intraday spread may represent a particularly interesting fast alpha for investors operating in the small- and micro-cap space**. Our results suggest that, historically, buying exposure near the close and reducing exposure near the following open would have been preferable to the opposite sequence. In practice, **investors who already intend to buy micro-cap exposure could consider using MOC orders, while those looking to reduce positions might benefit from waiting until the next session's open and using MOO orders instead**.

For more sophisticated market participants, the IWC results also suggest that part of the **effect may be directly tradable through a liquid ETF vehicle and in a long/short fashion**. Although transaction costs, slippage and market impact must be carefully considered (particularly as capital scales) the historical overnight-versus-intraday spread appears sufficiently large to deserve further investigation.

As is often the case in quantitative investing, discovering an anomaly is only the beginning. The more interesting challenge is understanding where that anomaly is strongest, how it interacts with other signals, and whether it can ultimately be transformed into a robust and implementable source of alpha.

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